

Selective Cross-Model Verification for Emotion Recognition in Conversations: When and Why Heterogeneous Review Helps

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Abstract

Emotion recognition in conversation (ERC) with fine-tuned large language models (LLMs) has reached strong accuracy, yet errors concentrate on a hard-to-detect minority of utterances. We introduce gated heterogeneous review, a deployment-time wrapper that augments an already trained ERC classifier without retraining it: a second, heterogeneous LLM serves as a selective verifier that is invoked only when the primary model is uncertain. On MELD (7-class) with eight independent seeds, gated heterogeneous review improves weighted-F1 by $+0.49 \pm 0.46$ percentage points (paired t : $p = 0.019$; bootstrap 95% CI $[+0.18, +0.77]$; Cohen’s $d_z = 1.07$) at an expected cost of only $1.27\times$ forward passes ($1.33\times$ in measured batch-1 wall-clock latency, 2.38 utterances/s on a single 48 GB GPU), while cheaper alternatives all fail: self-consistency sampling (three paired seeds, two temperatures; 11 of 12 configurations negative, -0.08 to -1.45), homogeneous self-debate (-0.64), and a weak 1.5B verifier (-0.05). A same-lineage Mistral-7B reviewer matched in standalone accuracy and routing rate, by contrast, yields no significant gain ($+0.18 \pm 0.52$, eight seeds, paired $p=0.37$), localising the effect to family-level independence rather than reviewer capability. The gain transfers to IEMOCAP ($+1.21 \pm 0.42$, three seeds, paired $p=0.037$) with adapters retrained on IEMOCAP and only the gate hyperparameters frozen from MELD. An eight-seed role reversal (Qwen2.5 proposes, LLaMA2 reviews) independently replicates the gain ($+0.71 \pm 0.42$, 8/8 seeds, paired $p=0.002$), with no significant difference between the two pairing directions. Stacking the gate on an independently trained InstructERC primary further shows the mechanism is portable across training pipelines, while its thresholds are not: generation-style confidence requires dev-level recalibration (two-threshold variant, $+1.33$). A frozen cross-register transfer to spontaneous DailyDialog chat does not replicate the gain, bounding the transfer claim to matched registers. Controlled analysis shows the benefit is multiplicative in verifier independence and capability, and concentrates in the high-coverage regime where abstention is disallowed.

Keywords: Emotion recognition in conversation, Large language models, Selective prediction, Model verification, Cost-aware inference

1. Introduction

Emotion recognition in conversation (ERC) is increasingly considered for high-volume applications such as contact-center quality inspection and affective monitoring (we treat these as illustrative deployment settings; our training corpora are scripted and acted transcripts, see Section 7), where errors on a small fraction of utterances, namely short, sarcastic, or context-dependent turns, carry disproportionate cost. Figure 1 illustrates the operating point we target: a fine-tuned LLM labels each utterance online, and its own label-sequence posterior already identifies where errors concentrate (accuracy rises monotonically from 0.48 in the lowest confidence bucket to 0.90 in the highest). The question is how to exploit this signal.

Our first finding is negative and constrains the design space: spending more compute on the same model does not help. Homogeneous self-debate reduces weighted-F1

by 0.64 points: two copies of one model agree on 91.3% of utterances, including their mistakes, and self-consistency sampling at $3\text{--}5\times$ budget degrades accuracy by $1.5\text{--}2.0$ points. Errors of a fine-tuned 7B classifier are correlated blind spots, not random noise; decorrelating them requires parameter-level independence. This motivates a second finding: an independently trained reviewer of comparable capability helps exactly when invoked selectively, while the same reviewer under a weaker gate or a weaker reviewer under the same gate both fail.

The interplay between these two findings has an important consequence for the design of LLM pipelines. The gain is not additive with the reviewer’s standalone accuracy: a reviewer that is weaker than the primary overall (68.67 vs. 68.88 weighted-F1, eight seeds) still improves the pipeline by $+0.49$ points, because its errors are decorrelated from the primary’s. Conversely, a stronger reviewer that is not independent (e.g., a second checkpoint of the same model fine-tuned on the same data) yields no gain despite higher standalone accuracy. The mechanism there-

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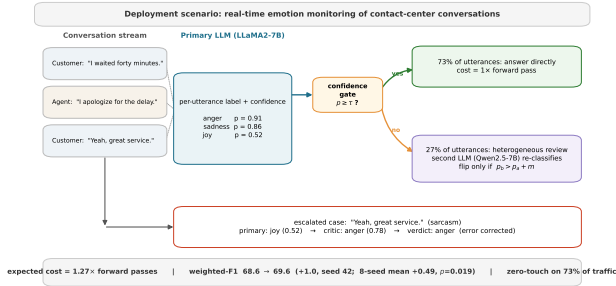


Figure 1: Application scenario. A primary LLM labels the conversation stream with calibrated confidences; only the low-confidence tail (27% of traffic) is escalated to a heterogeneous reviewer, correcting errors such as sarcastic anger misread as joy at an expected cost of 1.27x forward passes.

fore rewards a specific kind of diversity, meaning independence of error modes rather than raw capability, and the gate ensures that the diversity is consulted only where it is most likely to be useful.

Accordingly, this paper does not compete with accuracy-improving training methods such as retrieval augmentation or curriculum learning. Gated review is a deployment-time wrapper that can be stacked on top of any primary that already exposes a confidence score, including a stronger one; our experiments deliberately use an unenhanced fine-tuned primary so that the wrapper’s incremental effect is measured cleanly in paired comparisons. Research questions. We organise the study around four questions. RQ1 (efficacy): does selective heterogeneous review improve weighted-F1 over the primary model alone, across seeds, at a bounded expected cost? RQ2 (necessity): which ingredients of the mechanism are necessary: is same-source extra computation (self-debate, self-consistency, a weak reviewer) sufficient? RQ3 (generality): does the gain survive changes of reviewer family, of the pairing direction, of dataset, and of the primary’s training pipeline? RQ4 (mechanism): can the gain be attributed to measurable factors (routing quality, error independence, capability parity) rather than to a particular prompt or decoding trick?

Contributions.

- A selective cross-model verification framework for ERC: a confidence gate (τ) routes uncertain utterances to an independently trained heterogeneous reviewer, and a confidence-margin rule arbitrates disagreements (Section 3). The framework is a deployment-time wrapper: it does not modify or re-train the deployed primary, requires one extra fine-tuned 7B adapter, and its only test-time decision is a threshold comparison (Section 3).
- A systematic control study showing that same-source extra computation (self-debate, self-consistency, or a weak 1.5B reviewer) is ineffective or harmful on ERC, and that review gain is multiplicative in reviewer in-

dependence and capability (Sections 5.1–5.2).

- A cost–accuracy Pareto analysis (1.27x expected cost on the frontier), a learned-arbiter bound showing the rule gate nearly exhausts its feature family, zero-tuning gate transfer to IEMOCAP with dataset-retrained adapters, and a pipeline-portability test on an InstructERC primary showing the mechanism transfers but its thresholds require dev-level recalibration (Sections 5.3–5.4, 6).

The remainder of the paper is organised as follows. Section 2 reviews ERC, LLM verification, confidence estimation, and model cascades. Section 3 formalises the gated-review mechanism and decomposes its gain into three multiplicative factors. Section 4 describes the datasets, training protocol, and calibration procedure. Section 5 reports the main result, the control ladder, cross-family and cross-dataset generalisation, the cost–accuracy frontier, and per-class analysis. Section 6 interprets the mechanism through the lens of bias–variance and ensemble theory, derives practical deployment guidelines, and bounds the remaining oracle gap. Section 7 states limitations, and Section 8 concludes.

2. Related Work

2.1. Emotion recognition in conversation

Early ERC systems encode conversational context explicitly: recurrent and memory models track speaker-specific states [1], graph networks propagate affect over speaker and discourse relations [2, 3], and hierarchical or common-sense-enhanced transformers model global–local context [4, 5, 6]. Contrastive and label-refinement objectives further improve robustness on imbalanced labels [7, 8, 9]. A broader review of affective computing, including multimodal extensions, is provided elsewhere [10].

A recent generative shift reformulates ERC as instruction-following: fine-tuned LLMs with retrieval templates and emotion-alignment auxiliary tasks [11], or augmented with external commonsense and multimodal emotion knowledge [12], now outperform discriminative baselines by 3–6 weighted-F1 points on MELD. This shift raises a natural question: if a single generative model already encodes the task structure, can a second generative model add value beyond what more training data or a larger backbone provides? Our answer is conditional: the second model helps only when it is independently trained, comparably capable, and invoked selectively. Any of the classifiers above can serve as the primary model in our framework, and we deliberately keep the primary training objective (plain supervised fine-tuning) free of retrieval, external knowledge, or multimodal signals, so that the review mechanism is evaluated without confounding it with a stronger base system.

The discriminative-to-generative transition also changed the error profile. Discriminative models make errors that

are largely uncorrelated across architectures (different inductive biases), so simple voting ensembles were already effective [13]. Generative LLMs fine-tuned on the same data, even from different pre-trained families, make correlated errors because they share the same training distribution and the same label semantics. This correlation is why naive ensemble methods (majority vote, confidence-pick) yield diminishing returns on generative ERC, and why the gate must be selective—the value of the second model is concentrated on the instances where the first model is most likely to be wrong, not spread uniformly.

2.2. LLM debate, verification, and self-consistency

Multi-agent debate lets several LLM instances criticize and revise open-ended generations, improving factuality on reasoning and QA tasks [14]; self-consistency marginalizes over stochastic chain-of-thought samples [15], building on the observation that explicit reasoning traces improve LLM problem solving [16, 17]. Both mechanisms exploit diversity over long, open-ended outputs. We test whether the mechanism transfers to a setting where the output space is a single label token and a fine-tuned classifier’s errors are systematic rather than random: our homogeneous self-debate and self-consistency controls (at two sampling temperatures) both degrade weighted-F1 (Section 5.2), indicating that the benefit of deliberation depends on parameter-level error independence, which same-model sampling cannot provide. This also positions the work relative to LLM-as-a-verifier usage: a verifier helps only when its errors are decorrelated and its capability is matched, both of which we manipulate directly. Aggregating several reviewers’ signals in a reward-modelling context has also been studied [18]; our arbitration rule is a gated two-reviewer special case, and Section 6.4 shows that a learned arbiter over the same features does not beat the rule. Two recent ERC systems are closer neighbours. DARE [19] orchestrates a multi-turn adversarial debate between liberal and conservative agents instantiated by the same multimodal LLM over an open-vocabulary label set; our reviewers instead belong to heterogeneous, independently fine-tuned families, the label set is closed and short, and arbitration is a single gated call that generates no extra rationale. Lantern [20] prompts a frozen general-purpose LLM (GPT-4 or a 405B model) to reweight a discriminative ERC model’s class posteriors on IEMOCAP; our reviewer is itself a 7B model fine-tuned on the task, so the correction comes from an independent learned classifier rather than from frozen-model knowledge, at roughly one extra forward pass per triggered utterance rather than a large-model call.

2.3. Confidence estimation and uncertainty in LLMs

Using a model’s own uncertainty to decide when not to trust it has a long history in selective classification [21, 22, 23]: a classifier with a reject option trades coverage for risk, and its quality is summarised by the risk-coverage curve rather than by a single operating point. For

deep networks, the softmax response is a convenient but systematically overconfident score [24], and pre-trained transformers inherit this behaviour even after fine-tuning [25]. For LLMs, three confidence families have been studied [26, 27]: (i) logit-based scores derived from token probabilities of the answer span; (ii) verbalised scores elicited by prompting the model to state its own confidence [28, 29, 30]; and (iii) consistency-based scores that aggregate over multiple sampled generations [15, 31]. Our gate uses family (i)—the teacher-forced label-sequence posterior—which is the cheapest (no extra generation) and the only one directly comparable across the two models, but Section 5.1 shows it is miscalibrated in absolute value ($ECE \approx 0.1$) while remaining a usable ranking signal (Spearman ≈ 0.44). Our setting differs from classical selective prediction in one crucial respect: the rejected (low-confidence) instances are not abstained on but are re-answered by a second, independently trained model, so the relevant quantity is not the risk-coverage trade-off of a single model but the conditional accuracy of the reviewer on the routed subset. This also distinguishes the mechanism from consistency-based uncertainty (family iii): we show in Section 5.2 that sampling the same model fails to correct its errors, so the useful signal must come from parameter-level independence rather than from output variability.

2.4. Model cascades and selective prediction

Model cascades [32, 33] route queries to the cheapest model expected to answer correctly, solving $\arg \min$ cost subject to a quality constraint; selective prediction [22, 21] trades coverage for risk on a single model. Confidence-triggered escalation has likewise been formalised for LLM judging with provable human-agreement guarantees [34] and as hierarchical chains with multi-level abstention along an LLM intelligence hierarchy [35]. These mechanisms either escalate to a strictly stronger model or abstain; our second model is a comparable-capability verifier and the main branch never abstains (the abstention interaction is studied separately in Section 5.5). Our setting differs in both objective and mechanism: we $\arg \max$ accuracy subject to a bounded cost ($\leq 1.3\times$), and the second model is not a stronger fallback in a cascade but a comparable-capability, independently trained verifier whose value comes from error decorrelation—a cascade from a weaker to a stronger model cannot reproduce the effect, since our reviewer alone is weaker than the primary model (68.67 vs. 68.88, eight seeds) yet still helps. Conversely, escalation to a stronger reviewer would confound the gain with capability difference; our control ladder (Section 5.2) isolates exactly this factor.

3. Method

3.1. Problem formulation

Let $C = ((u_1, s_1), \dots, (u_n, s_n))$ be a conversation with utterance text u_i and speaker s_i , and let $\mathcal{L}$ be the fixed label

Algorithm 1 Selective cross-model verification at test time

Require: prompt x , labels $\mathcal{L}$, gate τ , margin m ; primary f_θ , reviewer g_ϕ

- 1: $(\hat{y}_a, p_a) \leftarrow f_\theta(x)$ ▷ greedy label + label posteriors
- 2: if $p_a \geq \tau$ then
- 3: return $\hat{y}_a$ ▷ shortcut: no second model
- 4: end if
- 5: $(\hat{y}_b, p_b) \leftarrow g_\phi(x)$
- 6: if $\hat{y}_b \neq \hat{y}_a$ and $p_b > p_a + m$ then
- 7: return $\hat{y}_b$ ▷ selective flip
- 8: else
- 9: return $\hat{y}_a$
- 10: end if

set ($|\mathcal{L}|=7$ on MELD, 4 on IEMOCAP). The task is to assign a label $y_i \in \mathcal{L}$ to every utterance from its speaker-tagged dialogue history. A fine-tuned primary model f_θ maps the prompted history to a label by greedy decoding, $\hat{y}_a$, and additionally assigns a label-sequence posterior $p_a(\ell)$ to every candidate label (Section 3.2). A second, independently trained model g_ϕ from a different model family produces $(\hat{y}_b, p_b)$ under the identical label set and prompt contract. Inference cost is measured in expected forward passes: one for f_θ , plus g_ϕ only when invoked.

3.2. Confidence estimation and gated invocation

Labels are generated as short strings (e.g., the token sequence “_neutral”), so we teacher-force the full label token sequence and define $z_i(\ell) = \sum_t \log p_\theta(z_t \mid x, z_{<t})$, $p_a(\ell) = \exp z_i(\ell) / \sum_{\ell' \in \mathcal{L}} \exp z_i(\ell')$, using the label posterior $p_a = p_a(\hat{y}_a)$ as the confidence signal. The leading space token is included for both models: LLaMA2 tokenizes some labels with two tokens while Qwen2.5 uses one, and stripping the first token would break cross-model parity. We do not use this scalar as an estimated probability—Section 5.1 shows it is miscalibrated—but only as a ranking signal for routing. Prompts are encoded without the tokenizer’s truncation and left-truncated with the BOS retained, preserving the target utterance and most recent history; the full prompt templates and truncation implementation are released with the code.

The gate invokes the reviewer only on the low-confidence tail, and a confidence-margin rule arbitrates disagreements (Algorithm 1). The gate threshold τ and margin m are calibrated once on the pooled development set and frozen for all test evaluation (Section 4).

3.3. When should gated review help? A gain decomposition

Before describing the reviewer, we make the mechanism’s requirements explicit with a simple decomposition that the experiments then instantiate factor by factor. Write T for the trigger event $\{p_a < \tau\}$ and D for the disagreement event $\{\hat{y}_b \neq \hat{y}_a\}$. A flip changes the prediction

only on $T \cap D$, so the expected accuracy gain of the gate decomposes as

$$\mathbb{E}[\Delta] = P(T) (P(\text{rescue}|T) - P(\text{harm}|T)), \quad (1)$$

where $\text{rescue} = \{\hat{y}_a \neq y, \hat{y}_b = y\}$ and $\text{harm} = \{\hat{y}_a = y, \hat{y}_b \neq y\}$. Each side factorises further along the mechanism’s causal chain:

$$P(\text{rescue}|T) = \underbrace{P(\hat{y}_a \neq y|T)}_{\text{routing}} \cdot \underbrace{P(D|\hat{y}_a \neq y, T)}_{\text{independence}} \cdot \underbrace{P(\hat{y}_b = y|D, \hat{y}_a \neq y, T)}_{\text{capability}}, \quad (2)$$

and analogously for harm with the roles of correct and wrong exchanged. Equation (2) exposes three multiplicative requirements, each of which can independently drive the gain to zero:

- **Routing quality:** the gate must enrich errors in the triggered bucket. On MELD the primary’s unconditional error rate is ≈ 0.31 , while conditioned on triggering it rises to 0.56 (Section 5.1)—a 1.8 $\times$ enrichment supplied by confidence ranking.
- **Independence:** the reviewer must disagree specifically on the primary’s errors. A homogeneous copy fails this factor: it agrees on 91.3% of utterances including the primary’s mistakes (Section 5.2), so $P(D \mid \hat{y}_a \neq y) \approx 0$ and Equation (1) collapses regardless of capability. Self-consistency sampling fails for the same reason by construction: samples from one parameter set share its systematic errors.
- **Capability:** conditional on disagreeing with a wrong primary, the reviewer must be right often enough that rescues outnumber harms. The 1.5B reviewer satisfies independence (56.1% agreement) but fails this factor—it degenerates toward the majority class—so the second and third factors cannot compensate for the first being irrelevant (Section 5.2).

The harm side of Equation (1) is governed by the same factors with the primary correct, which is why the margin m matters: requiring $p_b > p_a + m$ selectively suppresses flips where the reviewer’s confidence advantage is small, and these are precisely the flips whose rescue probability is lowest. The decomposition also predicts the ceiling effect observed in Section 5.1: as the primary improves, $P(\hat{y}_a \neq y \mid T)$ shrinks for any fixed ranking quality, so the expected gain attenuates—the strongest primary seed gains approximately zero. We return to Equation (2) in Section 6.1, where each factor is measured directly.

3.4. Heterogeneous reviewer with optional reasoning hint

The reviewer g_ϕ is a 7B model from a different family trained on the same SFT data, prompt contract, and epoch budget, so that family—not training protocol—is the manipulated source of independence. Its classification

Table 1: Dataset statistics (official splits; utterances).

Dataset	Train	Dev	Test	$ \mathcal{L} $
MELD	9,989	1,109	2,610	7
IEMOCAP	5,163	647	1,623	4

prompt optionally appends a short self-check hint listing common ERC confusions (literal wording vs. contextual affect, rare-class degeneration). We treat the hint strictly as an ablatable component and make no claim of benefit from it: the multi-seed ablation is mixed (Section 5.2), and the hint-free configuration is the simpler default.

4. Experimental Setup

Datasets. We use the official splits of two English conversational corpora (Table 1). MELD [36] contains multi-party TV-sitcom dialogues with seven emotions; IEMOCAP [37] contains two-speaker acted sessions, evaluated on four emotion classes under the standard mapping. Both are scripted or performed speech domains—implications for deployment-domain transfer are stated in Section 7. Only the text modality is used; MELD’s audio and video streams are deliberately excluded so the mechanism is not confounded with multimodal cues.

Models and training. The primary model is LLaMA2-7B [38] with LoRA [39] (rank 32; 33.6M trainable parameters), and the main reviewer is Qwen2.5-7B [40] with LoRA rank 64 (LoRA $\alpha=128$, dropout 0.05, q/v -projections, learning rate 2×10^{-4} , effective batch size 32); a Mistral-7B [41] reviewer is trained under the same protocol as a second-family control. The reviewer uses a higher LoRA rank than the primary (64 vs. 32); this asymmetry is deliberate, giving the reviewer ample capacity, and it does not drive the gain: despite twice the trainable parameters the reviewer’s standalone weighted-F1 (68.67) does not exceed the primary’s (68.88) over the eight seeds, while the family comparison in Section 5.3 holds reviewer rank fixed at 64. All models are fine-tuned for three epochs on identical SFT data with the same prompt contract. Inference is greedy decoding with `max_new_tokens=8` for labels; confidence scoring uses teacher-forced label sequences as defined in Section 3.2.

The SFT prompt presents the speaker-tagged history and asks for the emotion of the final utterance as a single label from $\mathcal{L}$; no retrieval, external knowledge, or chain-of-thought is used at any stage, by design, so that the review mechanism is evaluated against the weakest reasonable base system rather than confounded with training-time enhancements. Training one adapter takes approximately 2.1 h (primary) and 2.6 h (reviewer) on a single GPU, and the eight-seed study therefore varies only the LoRA initialisation and data order, holding data, prompts, and hyperparameters fixed. We deliberately use LoRA rather than full fine-tuning: besides fitting on one GPU, it makes the

independence manipulation clean, since the two reviewers share no trainable parameters with the primary and differ only in their frozen backbones. The Appendix lists the exact templates, the left-truncation rule, and the label strings.

Calibration protocol. The pair (τ, m) is selected by a single grid search over the pooled development predictions of three seeds (3,327 MELD utterances), maximizing dev weighted-F1, yielding $\tau=0.65$, $m=0.05$. Test data participate in neither threshold selection nor prompt modification; all eight test seeds and the entire IEMOCAP evaluation use these frozen hyperparameters.

Seeds, metrics, and hardware. We report eight paired seeds (42–49) on MELD and three on IEMOCAP, varying initialization of both models. The primary metric is weighted-F1; we also report macro-F1 and accuracy. Statistical evidence uses the paired t -test, Wilcoxon signed-rank test, McNemar’s test [42], and a paired bootstrap over seeds (100k resamples). The confirmatory family, defined before any test-set evaluation with the analysis protocol archived in the released code, consists of the paired t -test and the Wilcoxon signed-rank test on the eight-seed weighted-F1 gain on MELD; Holm correction within this two-test family leaves both adjusted $p < 0.05$ (0.039 and 0.039). Exact sign tests and per-seed McNemar tests are reported alongside as effect-robustness evidence rather than as separate confirmatory hypotheses. All other reported p -values—the hint ablation, IEMOCAP gate variants, self-consistency grid cells, and per-class analyses—are exploratory and are not multiplicity-controlled; we report effect sizes and seed distributions for these rather than relying on significance alone. Two 7B models co-reside on one 48 GB GPU (NVIDIA RTX PRO 5000, bf16). A direct batch-1 benchmark of the production inference path (greedy 8-token answer plus the seven-label confidence scoring, 400 timed test utterances after warm-up) measures 28.3 GB peak allocated memory (34 GB reserved per `nvidia-smi`) and per-utterance latency of 317 ms for the primary alone (P50 320, P95 378; 3.16 utterances/s), 419 ms on average under the frozen gate (P50 330, P95 757; 2.38 utterances/s), and 697 ms under unconditional review (1.43/s); the confidence pass is one batched teacher-forced call over the seven candidates. In token terms the mechanism is cheap: the label answer is at most 8 generated tokens (Appendix A), and the confidence score is one teacher-forced pass over each of the 7 candidate label sequences (2–3 scored tokens each), so a triggered review adds one critic forward of the same profile rather than a long rationale. Under the frozen gate the reviewer fires on 26.5% of utterances; the expected cost is $1.27\times$ in forward-pass accounting, and the measured wall-clock ratio is $1.33\times$ —slightly higher because one critic call is about $1.2\times$ slower than a primary call in our model pair, but far below the $2\times$ of unconditional review. The unit of the cost figures throughout is one batched model call over a

Table 2: Eight-seed main results on MELD test (weighted-F1, %).

Seed	Base	Ours	Δ
42	68.60	69.63	+1.04
43	68.97	68.70	-0.27
44	68.76	69.66	+0.90
45	69.14	69.57	+0.43
46	68.83	69.38	+0.56
47	69.78	69.68	-0.10
48	68.35	69.09	+0.74
49	68.61	69.25	+0.65
Mean	68.88	69.37	+0.49 $\pm$ 0.46

prompt of the same profile (the seven-label teacher-forced confidence batch counts as one such call); this confidence pass is paid in every condition, including the primary-only baseline, so it cancels in the $1.27\times$ ratio, and the only marginal cost of gating is the extra reviewer call on the triggered subset.

All reported numbers are computed on the official test splits with the frozen gate ($\tau=0.65$, margin 0.05) and the frozen adapters; no hyperparameter is tuned on test data. The eight seeds differ only in the LoRA weight initialization and the data shuffling order of SFT, not in the data, prompts, or hyperparameters. Paired tests are appropriate because the two conditions (base vs. gated) are evaluated on the same utterances with the same primary adapter, so the pairing controls for both utterance-level difficulty and seed-level variation. We report all seeds, including the two negative ones (43 and 47), and we report the full distribution rather than cherry-picked maxima. The per-seed gains and the aggregate statistics are reproducible from the released records.

5. Results

5.1. Main result: eight-seed paired evaluation

The paired mean gain is $+0.49 \pm 0.46$ weighted-F1 points (six positive, two negative seeds): paired $t=3.030$, $p=0.019$; paired bootstrap 95% CI $[+0.18, +0.77]$; Wilcoxon $p=0.039$; Cohen’s $d_z=1.07$. The result is not driven by the best seed—excluding seed 42 leaves $p=0.046$. We retain both negative seeds in all aggregates; seed 47 pairs the strongest primary model with a near-zero gain, consistent with a diminishing-returns ceiling. This effect is independently replicated by a confirmatory eight-seed role-reversal pairing (Qwen primary, LLaMA2 reviewer; $+0.71 \pm 0.42$, 8/8 seeds positive, $p=0.002$), with no significant difference between pairing directions; we retain this LLaMA2-primary pairing as the main result because it aligns with the published backbone anchors, and report the replication in Section 5.3.

Position relative to published systems.. Table 3 places our numbers against reported MELD results. We make

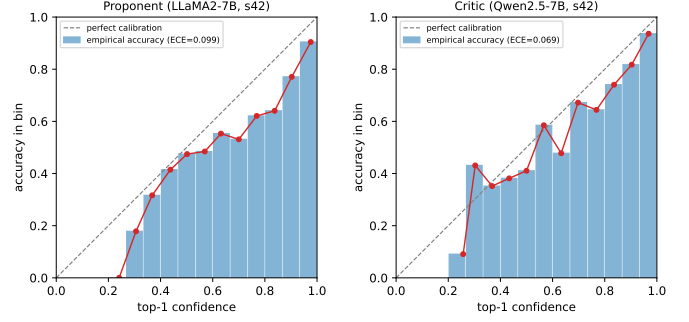


Figure 2: Reliability diagrams (seed 42). Both models are overconfident, but confidence still orders utterances by correctness (Spearman ≈ 0.44); the gate uses the ranking, not the absolute values.

no state-of-the-art claim: the strongest recent text-only system is PRC-Emo at 70.44 (AAAI 2026, prompt enrichment plus a dedicated retrieval repository plus curriculum learning on Qwen3-8B), while systems that augment text with external knowledge (speaker biography, video emotion knowledge) reach 69.8–71.9 and multimodal audio-visual systems reach 66.7–69.9. Two observations bound our claim. First, the plain-LoRA ablation reported by PRC-Emo itself is 68.72, and its released-pipeline re-runs—together with our reproduction of InstructERC’s released pipeline (66.29; the 69.15 entry point is absent from its public repository, as detailed in the table footnote)—place unenhanced 7–8B generative fine-tuning in the 66–69 band that our own plain-SFT primary (68.88 ± 0.41) occupies; the reported 70.44 level is reached through training-time enhancements that are complementary to our mechanism. Second, our positioning rests on reported water levels plus our own paired multi-seed numbers, never on cross-implementation significance. Gated review enters the 69.3+ band at $1.27\times$ expected cost; the claim is an orthogonal deployment-time mechanism that can wrap any stronger primary model, including systems with the PRC-Emo-style enhancements, rather than a stronger single classifier.

Calibration of the gating signal.. Because the gate consumes p_a , we verify what the confidence can and cannot be trusted for. Both models are systematically overconfident (Figure 2): expected calibration error (ECE-15; 24) is 0.097 ± 0.006 for the primary and 0.070 ± 0.003 for the reviewer over three seeds, with a confidence–accuracy gap of $+0.10$ and $+0.07$. Absolute confidences must therefore not be read as probabilities. The ranking, however, is stable and useful: Spearman correlation between confidence and correctness is 0.440 ± 0.006 (and the risk–coverage analysis gives AURC 0.156 vs. 0.307 for random ranking, Section 5.5). An isotonic mapper fit on dev reduces test ECE to 0.025/0.019 while leaving the ranking virtually unchanged, so the gate is robust to the calibration error: it ranks, rather than interprets, the scalar.

The decoupling between calibration and ranking has a practical consequence for the gate design. If the confi-

Table 3: Position on MELD test (weighted-F1, %). External values are as reported by their papers under the official test split; they are not paired reproductions (preprocessing, context windows, and seed counts differ), so no significance is attached to cross-row differences. Text-only and multimodal results are separated; only the text-only block is comparable in input modality.

Paradigm	System	Year	W-F1
Discriminative	DialogueRNN [1]	2019	57.0
Discriminative	DialogueGCN [2]	2019	58.1
Discriminative	DialogXL [4]	2021	62.4
Discriminative	DAG-ERC [3]	2021	63.1
Discriminative	COSMIC [5]	2022	64.2–65.2
Discriminative	EmotionFlow-large [6]	2022	66.5
Discriminative	SACL-LSTM [†] [7]	2023	66.9
Discriminative	HiDialog [8]	2023	67.0
Discriminative	EmoTrans (RoBERTa-large) [9]	2024	68.0
Generative 7B	InstructERC [§] [11]	2024	69.15
Generative 7B	InstructERC, released pipeline reproduced by us	2026	66.29
Generative LLM	CKERC [43]	2024	69.27
Generative LLM	BiosERC (speaker-biography knowledge) [44]	2024	69.83
Generative 8B	PRC-Emo (full P+R+C) [45]	2026	70.44
This work	Primary SFT, 8-seed mean	—	68.88±0.41
This work	Gated review, 8-seed mean (1.27× cost)	—	69.37±0.39
Multimodal	M2FNet [46]	2022	66.7
Multimodal	TelME [47]	2024	67.4
Multimodal	ELR-GNN [48]	2024	69.9
Multimodal	DialogueLLM-7B (video knowledge) [12]	2024	71.90

[†]Reported for a single seed. [§]The 69.15 result uses the authors’ unified-label training recipe (MELD+IEMOCAP+EmoryNLP mixed for 8 epochs) whose entry script (main_Unilabel.py) is absent from the public repository (verified 2026-10-04). Under the only released pipeline (MELD-only, 10-epoch LoRA, seed 42) we reproduce 66.29; the paired same-backbone anchor in this paper is therefore our own primary

SFT (68.88), trained and evaluated under one unified protocol. Inference cost is not compared across rows: none of the listed systems reports latency or hardware, CKERC and BiosERC additionally invoke external LLM APIs, and the multimodal block incurs audio/video encoding; the controlled cost comparison is instead given as a cost–accuracy Pareto under one protocol in Fig. 3.

dence were perfectly calibrated, one could set τ by reasoning about absolute error rates (e.g., trigger when the model is more likely wrong than right, $\tau=0.5$). Because the confidence is overconfident, that reasoning fails: the model reports $p_a=0.8$ on utterances where its accuracy is only 0.7, so a naively chosen τ under-triggers. Our grid search over dev finds the operating point empirically, bypassing the need for calibrated absolute values. This is also why the margin m is expressed as a difference of confidences rather than a ratio: a ratio would be sensitive to the absolute scale, whereas a difference of two scores from the same (miscalibrated) family preserves the ordering information that the gate actually uses.

5.2. Cheaper alternatives all fail, and the reasoning hint is not essential

We hold the primary model and test set fixed and compare against cheaper or structurally similar alternatives. Homogeneous self-debate (a second LLaMA2-7B copy as reviewer) changes weighted-F1 by -0.64 : the two copies agree on 91.3% of utterances, including their mistakes, so extra compute reproduces rather than corrects the errors. Self-consistency sampling of the primary model (top- p 0.9, majority vote of 3/5 samples, ties broken by label posteriors) was evaluated as a strictly paired three-seed control (seeds 42–44, same-seed greedy baseline, $n=2610$ each)

at two temperatures. Mean weighted-F1 changes versus same-seed greedy are uniformly negative: -1.45 ± 0.57 (SC@3, $T=0.7$, paired $p=0.048$, 3/3 seeds negative) and -0.98 ± 0.59 (SC@5) at $T=0.7$, narrowing to -0.33 ± 0.30 and -0.08 ± 0.27 at $T=0.3$ (11 of 12 seed×configuration cells negative; the single positive cell is $+0.17$ at $5\times$ cost). Lower temperature halves the damage—part of the high-temperature loss is sampling noise—but as temperature approaches greedy decoding, SC merely becomes harmless rather than beneficial: the best SC variant (68.70 at $5\times$ cost) still matches rather than beats the same-seed greedy baseline (68.78 at $1\times$; 68.88 over eight seeds) and trails gated review (69.33 over the same three seeds, 69.37 over eight) at $1.27\times$. The failure is structural rather than temperature-driven: same-parameter samples share the classifier’s correlated blind spots on a short-label task. A weak 1.5B reviewer has genuine independence (56.1% agreement) but no parity in capability (it degenerates to a majority-class predictor, W-F1 0.486), and yields -0.05 . Finally, full-cost ensembles at 100% reviewer invocation reach 69.49 (confidence-pick) and 69.55 (flip on every disagreement), versus our gated 69.37 at 27.0% invocation: four times the review budget buys at most 0.18 points. Together these controls show the gain requires reviewer independence and capability parity and selective invocation jointly; Fig. 3 places all these controls on one cost–

accuracy plane.

Ablation: reasoning hint (mixed result reported).. The reviewer prompt optionally includes a short self-check hint. Removing it on the same adapters gives $+0.84$, $+0.59$, $+0.73$ across seeds 42–44 (mean $+0.72 \pm 0.13$, all positive, $p=0.010$), versus $+1.04$, -0.27 , $+0.90$ with the hint (mean $+0.56 \pm 0.72$). The hint helps marginally in two seeds but materially hurts in seed 43 (-0.85); its average effect is not distinguishable from zero and slightly below the hint-free variant. We therefore make no claim that the hint contributes: the architecture (gating plus heterogeneous review) carries the gain, and the hint-free configuration is arguably the safer deployment choice. We report this negative component result explicitly rather than suppressing it.

5.3. Cross-family and cross-dataset generalization

Second reviewer family: family independence as the binding constraint. Does review help whenever the second model is a capable 7B classifier trained under the same protocol, or does it require family-level independence? We replace the Qwen2.5-7B reviewer with a Mistral-7B reviewer—same SFT data, prompt contract, and epoch/LoRA budgets—in eight paired runs (seeds 42–49), with the LLaMA2-7B primary and frozen gate unchanged. Routing is matched by construction: the gate invokes the Mistral reviewer on 26.5% of test utterances versus 26.5% for the Qwen reviewer. Reviewer capability is matched as well: Mistral’s standalone weighted-F1 is 68.82 ± 0.45 versus 68.67 ± 0.32 for Qwen (eight seeds), so the earlier single-seed observation of a weaker Mistral reviewer does not survive reseeding. The gated gains, however, do not replicate: $+0.18 \pm 0.52$ points ($68.88 \rightarrow 69.05$), six of eight seeds positive ($+0.29$, $+0.06$, $+0.43$, -0.82 , $+0.29$, -0.26 , $+0.88$, $+0.53$), and no confirmatory test rejects zero (paired $t=0.95$, $p=0.37$; exact Wilcoxon $p=0.25$; exact sign $p=0.29$; aggregate rescue/harm $695/543 = 1.28:1$). This is the decomposition of Equation (2) tested in situ: with capability and routing held constant, a reviewer from the LLaMA architectural lineage—plausibly drawing from the same parameter-level error pool as the LLaMA2 primary—fails to convert escalations into net corrections, whereas the unrelated Qwen family does ($+0.49 \pm 0.46$, $p=0.019$). The 0.31-point difference between the two reviewer families is itself not significant (Welch $t=1.28$, $p=0.11$), so we read the Mistral control as evidence that family-level independence is necessary for the gain, not as a quantified lineage effect. A fully factorial (capability $\times$ independence) sweep with tokenizer- and lineage-matched controls would tighten the attribution and is left to future work.

Role-reversal symmetry: an eight-seed independent replication.. The main pairing fixes LLaMA2-7B as primary and Qwen2.5-7B as reviewer, so the gain could in principle depend on the direction of the pairing rather than on the mechanism. We test this with a full replication in

the reverse direction: Qwen2.5-7B proposes and LLaMA2-7B reviews, across eight new paired runs (seeds 42–49), each family using its own adapter trained under the same SFT protocol, with the same frozen gate ($\tau=0.65$, margin 0.05). The result independently reproduces the main effect: the primary alone scores 68.67 ± 0.22 and gated review reaches 69.38 ± 0.41 , a paired gain of $+0.71 \pm 0.42$ weighted-F1 points with all eight seeds positive ($+1.03$, $+1.20$, $+0.98$, $+1.04$, $+0.32$, $+0.70$, $+0.24$, $+0.14$); paired $t=4.78$, $p=0.002$, Wilcoxon $p=0.008$, exact sign test $p=0.008$, Cohen’s $d_z=1.69$. Review is invoked on $24.6 \pm 1.2\%$ of utterances with 593 rescues versus 393 harms (1.51:1), closely matching the main direction’s routing pattern (26.5% invocation; 525/414 pooled over eight seeds, 1.27:1).

Two features make this a genuine replication rather than a second look at the same result. First, the two directions do not differ in effect size: a seed-by-seed paired comparison of the gains gives a mean difference of $+0.21$ points with $t=0.86$, $p=0.42$, and the two gated systems end at virtually the same level (69.37 and 69.38); the larger mean gain in the reverse direction reflects its slightly weaker primary (68.67 vs. 68.88), i.e. more headroom, not a better system. Second, the reverse direction turns both negative seeds of the main study positive (seed 43: $-0.27 \rightarrow +1.20$; seed 47: $-0.10 \rightarrow +0.70$), indicating the occasional negative outcome is seed-specific noise rather than a direction-dependent failure of the mechanism. We deliberately retain the LLaMA2-primary pairing as the main result—it is the pairing aligned with the published backbone anchors in Table 3—and present the reversed pairing as a confirmatory independent replication, with its analysis plan fixed before its test evaluation. Because both directions share the same 2,610 test utterances, they are not independent samples and we do not pool their significance tests; the evidential claim is replication across pairing directions with paired tests within each.

Different primary training pipeline (InstructERC).. The strongest portability test is a primary whose training pipeline we did not design. We stack the gate on our reproduction of InstructERC’s released single-dataset pipeline [11] (LLaMA2-7B, LoRA $r=16$, 10 epochs; 66.47 W-F1, matching the 66.29 reference; Table 2), keeping the Qwen2.5-7B reviewer and its records unchanged. This primary differs from ours in both prompt format and training objective—it generates the label sequence rather than scoring candidate labels—so its confidence is a generation probability (mean per-token probability of the emitted label span) rather than a candidate-label posterior. The semantic mismatch is measurable and consequential: generation confidence concentrates near one (median 0.9999; only 2.7% of test utterances fall below the frozen $\tau=0.65$), so the zero-tuning gate fires on a sliver of the data and the gain collapses ($66.47 \rightarrow 66.58$, $+0.11$; rescue/harm 8/5). Two repairs give opposite outcomes, and the contrast is instructive. Recalibrating only the trigger τ on dev while keeping the margin comparison $p_b > p_a + 0.05$ still fails ($+0.05$;

6.1% trigger; rescue/harm 11/10): the margin compares confidences across scales, and when $p_a \approx 1$ the condition is almost never satisfiable. Replacing the margin by a per-reviewer threshold—flip iff $p_a < \tau_a$, $y_b \neq y_a$, and $p_b > \tau_b$, with both thresholds grid-searched on dev only ($\tau_a=0.995$, $\tau_b=0.60$)—restores the gain on test: 67.79 (+1.33), 23.8% trigger rate, rescue/harm 70/35 (2:1), at 1.24× expected cost, comparable to the main pairing’s +1.04 at 1.27×. The mechanism—low-confidence routing, independent review, disagreement filtering—therefore transfers across primary training pipelines, but the gate parameters do not: they encode the primary’s confidence semantics and must be recalibrated on dev for each new primary, and the arbitration must compare each model against its own threshold rather than across scales. We accordingly soften the portability claim: the gate is portable as a mechanism, not as a frozen configuration.

Cross-dataset transfer.. On IEMOCAP (4-class), both adapters are retrained on the IEMOCAP train split under the identical SFT protocol; only the gate hyperparameters are transferred, frozen from MELD ($\tau=0.65$, margin 0.05, zero test-time tuning). This zero-tuning gate transfer improves weighted-F1 by $+1.21 \pm 0.42$ points across three seeds (80.16 $\rightarrow$ 81.37; +0.92, +1.69, +1.02; all three positive; paired t , $p=0.037$), larger than IEMOCAP’s own dev-calibrated variant ($+0.40 \pm 0.05$, $p=0.005$). The frozen gate is better precisely because it is estimated from 3×1109 MELD dev utterances, whereas per-seed IEMOCAP calibration on only 647 dev utterances is noisy—two of three seeds select gates with a 2–3% trigger rate that leave most gains unexploited. Reviewer routing transfers as well: within the triggered bucket (13.6% of utterances), primary-model accuracy is only 49.6% and review lifts it to 56.5%, mirroring the MELD mechanism.

The asymmetry between the two transfer directions is instructive. The frozen MELD gate transfers to IEMOCAP because IEMOCAP’s 4-class label space is a subset of MELD’s 7 classes, so the primary’s confidence distribution on IEMOCAP is a coarse-grained version of its MELD distribution; the gate’s ranking of “hard” vs. “easy” utterances is preserved under this coarsening. The reverse direction would not work: a gate calibrated on IEMOCAP’s 4-class distribution would face 7-class MELD utterances with no support for the three extra classes, and its ranking would be undefined there. This is a general property of the mechanism, not a quirk of these two datasets: the gate inherits the label set of the primary it gates, so cross-dataset transfer is only well-defined when the target label set is a (possibly non-injective) image of the source label set. We return to this asymmetry in Section 7.

Out-of-domain transfer (DailyDialog): preliminary evidence.. To probe transfer beyond scripted/acted corpora, we apply the unchanged MELD-trained models and the frozen rule ($\tau=0.65$, margin 0.05, no target-domain tuning) to the 7,740-utterance DailyDialog test split, whose label

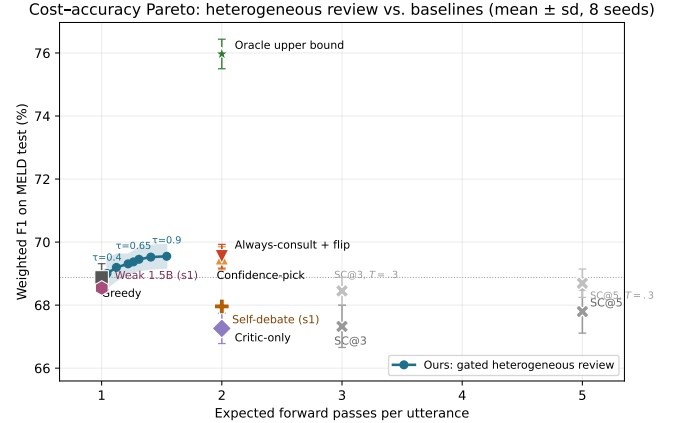


Figure 3: Cost-accuracy Pareto on MELD test (mean $\pm$ sd over eight paired seeds; single-run controls marked s1). Sweeping the gate τ traces the frontier, and the frozen operating point $\tau=0.65$ sits on it at 1.27× expected cost and 69.37 weighted-F1. Full review at 2× matches the gate only after quadrupling the review budget (at most +0.18), whereas homogeneous self-debate (2×, s1), self-consistency at both temperatures (3–5×), and a weak 1.5B reviewer ($\approx 1\times$, s1) do not improve over the greedy primary. The oracle marks the remaining arbitration gap bounded in Section 6.2.

set is mapped onto the seven MELD classes (Appendix B). We report two decoding protocols for transparency: logprob-argmax (argmax over teacher-forced candidate-label scores, the same decoding family from which the gating confidence p_a is extracted; used for the gated result below) and the greedy generation protocol used elsewhere in this paper, given as a reference (Appendix B). The primary model transfers well on its own (logprob-argmax weighted-F1 0.8236; greedy 0.7913), but the gated mechanism does not transfer its benefit: the MELD-trained reviewer is weaker on this distribution (0.7895), and gating lowers weighted-F1 to 0.8155 ($\Delta = -0.81$; 136 rescues vs. 304 harms), while invoking the reviewer on 14.97% of utterances (6.56% adopted) at 1.15× expected cost. DailyDialog is spontaneous chat heavily dominated by neutral utterances rather than balanced acted emotion, so both the confidence calibration that routes errors and the reviewer’s relative strength shift out of domain. We emphasise that this is preliminary, single-seed evidence and make no significance claim, but we report it in full because it suggests a boundary: the frozen gate generalises across datasets within the same register (MELD $\rightarrow$ IEMOCAP) but may not generalise across registers without target-domain calibration of the gate and a reviewer verified to remain comparably capable.

5.4. Cost-accuracy Pareto

5.5. Interaction with abstention

5.6. Case study

Table 5 shows three representative flips under the deployed rule ($\tau=0.65$, margin 0.05). In R1 the reviewer separates physiological disgust from the angry wording

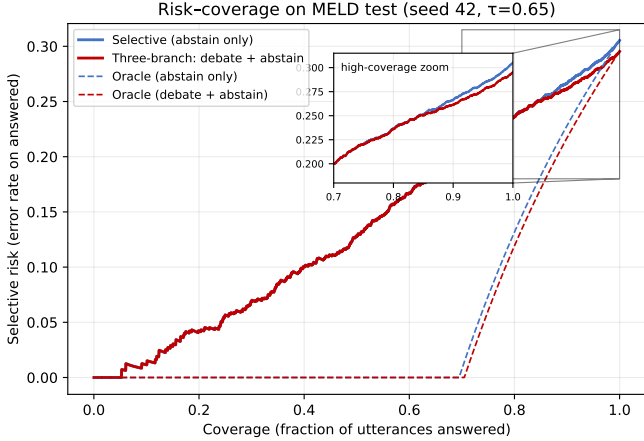


Figure 4: Risk-coverage with an abstention option (seed 42). The review gain concentrates in the high-coverage regime (inset) where abstention is disallowed; the AURC difference is not significant (CI crosses zero), so we report the review step as a high-coverage correction mechanism, not a substitute for abstention.

that misled the primary model, mirroring the cleanest per-class gain (disgust, pooled rescue/harm = 31/9 over eight seeds). In R2 both models are unconfident ($p_a=0.32$), but escalation separates a resigned sadness from literal neutrality. R3 captures suppressed annoyance behind a suggestion-form sentence. H1 is the failure mode we report openly: polite-but-distant phrasing is read as anger by the reviewer, and the margin rule fails to block the flip. The fear pair (F1/F2) shows both directions on a 50-instance class: a one-word cry for help is rescued, while panicked repetition about a lost heirloom is misread as anger with high reviewer confidence—evidence for routing very-low-confidence minority classes to abstention rather than forced arbitration.

The case study reveals a structural pattern in the mechanism’s successes and failures. Successful rescues (R1–R3) share a common structure: the primary’s error is driven by a surface lexical cue (angry wording, smoking, suggestion form) that overrides a weaker but decisive contextual cue (physiological disgust, resignation, suppressed annoyance). The reviewer, with its independent training, is less sensitive to the surface cue and recovers the contextual one. The harm case (H1) has the opposite structure: the reviewer invents a negative affect (anger) that is not present in the primary’s reading, and the margin rule cannot distinguish this from a genuine rescue because the reviewer’s confidence is high. This asymmetry—rescues recover suppressed context, harms invent spurious affect—suggests that the mechanism is most trustworthy when the primary’s low confidence is caused by cue conflict rather than by absence of signal.

5.7. Per-class analysis across seeds

Table 4 aggregates rescue and harm counts across all eight seeds on MELD test. The gain is not uniform: disgust (rescue/harm ratio 3.44), sadness (1.62), and neutral

Table 4: Per-class rescue/harm aggregated across 8 seeds on MELD test (8×2,610 repeated measurements, not independent samples). R/H ratio = rescues / harms; > 1 means net benefit.

Class	Total	Rescue	Harm	R/H ratio
neutral	10048	209	134	1.56
joyful	3216	73	71	1.03
sadness	1664	63	39	1.62
anger	2760	103	65	1.58
surprise	2248	38	77	0.49
disgust	544	31	9	3.44
fear	400	8	19	0.42
Total	20880	525	414	1.27

(1.56) benefit most, while surprise (0.49) and fear (0.42) are net negative. This asymmetry has two implications. First, the mechanism is most reliable on classes where contextual cues are strong but lexical cues are ambiguous (disgust masked as anger, sadness masked as neutrality). Second, the two net-negative classes share a property: they are the two smallest classes in MELD (fear: 50 test instances; surprise: 281), where both models are least reliable and the reviewer’s independent errors compound rather than correct the primary’s. Because these counts aggregate eight repeated measurements on the same test instances, a seed-level bootstrap of the pooled R/H ratio gives a 95% CI of [0.17, 0.75] for fear and [0.34, 0.69] for surprise, confirming the net-negative trend is stable across seeds despite the small class size. This motivates the abstention branch for rare classes (Section 5.5) and suggests that the gate threshold should be class-conditional in future work.

A closer look at the surprise row reveals the mechanism’s blind spot. Of the 77 surprise harms, 37 are flips from surprise to anger and 27 from surprise to neutral (the remaining 13 go to joyful or sadness)—the reviewer reads surprising utterances (short exclamations, abrupt topic shifts) as anger or as neutral, with high confidence, and the margin rule admits the flip because p_b is high. The reviewer’s confidence is miscalibrated on exactly this class: it is trained on the same MELD data, so it inherits the same lexical ambiguity between surprise and anger/neutral, and its independent training does not break the correlation. This is a systematic rescue-to-harm conversion that is not present on the other confusable pairs (disgust/anger, sadness/neutral), where the reviewer’s independent bias is less correlated with the primary’s. The pattern suggests that the margin rule should be class-pair-conditional: the threshold for flipping into class ℓ should depend on the primary’s confusion rate with ℓ , not just on the reviewer’s confidence. We leave this extension to future work.

5.8. Ablation summary

Table 6 consolidates all ablations and controls on MELD. The heterogeneous gate is the only configuration

Table 5: Case study on MELD test (seed 42). Bold: target utterance.

ID	Context → target	gold / primary / reviewer
R1	...“Ill fire him today and you go out with him for another week.” → “Are you kidding?! Another week with that sip, Ill kill”	disgust / anger (.55) / disgust (.86) ✓
R2	...“Then why are you smoking?” → “Well its very unsettling.”	sadness / neutral (.32) / sadness (.38) ✓
R3	...(sarcastic tip remark) → “would you just see my chiropractor, already.”	anger / neutral (.53) / anger (.62) ✓
H1	...“could you pass the cheese?” → “Listen uh, Id prefer it if you didnt call me Joey.”	neutral / neutral (.59) / anger (.68) ×
F1	...“The charitys on fire!” → “Help!”	fear / anger (.64) / fear (.81) ✓
F2	...“which one of you guys is Gunther...” → “Wheres my ring? My dead grandmothers wedding ring?”	fear / fear (.55) / anger (.83) ×

Table 6: Ablations and controls on MELD test. Gains in weighted-F1 (pp), paired over seeds where applicable. Single-seed results marked (s1).

Configuration	$\Delta W\text{-F1}$	Seeds	Significance
Heterogeneous gate (Qwen review)	$+0.49 \pm 0.46$	8	$p=0.019$
Same gate, no hint	$+0.72 \pm 0.13$	3	$p=0.016$
Same gate, with hint	$+0.56 \pm 0.72$	3	n.s.
Homogeneous self-debate	-0.64	1 (s1)	n.s.
Self-consistency SC@3, $T=0.7$	-1.45 ± 0.57	3	$p=0.048$
Self-consistency SC@5, $T=0.7$	-0.98 ± 0.59	3	n.s.
Self-consistency SC@3, $T=0.3$	-0.33 ± 0.30	3	n.s.
Self-consistency SC@5, $T=0.3$	-0.08 ± 0.27	3	n.s.
Weak 1.5B reviewer	-0.05	1 (s1)	n.s.
Full review (confidence-pick)	$+0.00$	1 (s1)	n.s.

that yields a positive, statistically significant gain; every cheaper or structurally similar alternative is either negative or indistinguishable from zero. The hint ablation is mixed (+0.72 without hint vs. +0.56 with hint, three seeds each), so the simpler hint-free configuration is the safer default. The 3×3 proponent–reviewer factorial (Section 6.1) confirms that independence and capability are jointly necessary.

The ablation table also exposes a design principle that is easy to miss in the per-experiment prose: the cost–accuracy trade-off is not monotone in the review budget. Self-consistency at 3–5× cost degrades accuracy; full review at 2× cost matches the gate at best; and the gate at 1.27× cost dominates both. The mechanism’s value comes not from the amount of compute but from where the compute is spent: selectively on the instances where the primary is most likely to be wrong, with an independently trained reviewer whose errors are decorrelated from the primary’s. Spending the same compute indiscriminately (full review) or on the same model (self-consistency) wastes the budget on instances that are already correct or on errors that are correlated with the primary’s.

6. Discussion

6.1. Mechanism: why heterogeneous review works

Three evidence-backed findings explain the gain. First, errors of one fine-tuned model are systematically correlated with themselves but partially decorrelated across families: two LLaMA2 copies agree on 91.3% of utterances—including their mistakes—while the heterogeneous pair agrees on 79.9%, which is exactly the slack a verifier can exploit. Second, independence and capability are jointly necessary rather than alternatives: the 3×3 proponent–reviewer factorial is positive in 8 of 9 cells, all six off-diagonal pairings are positive, and the sole negative cell pairs the weakest proponent with the weakest reviewer; a 1.5B reviewer with high independence but no capability yields nothing (Section 5.2). Third, the errors that can be corrected are routable: the primary model’s confidence orders them (Section 5.1), and rescues concentrate on mid-frequency confusable classes (disgust rescue/harm = 15/2, anger 49/14) rather than on the rarest classes, where both models fail. The ceiling observation complements this picture—the strongest primary seed gains approximately zero—so expected value is largest for ordinary, non-saturated base classifiers.

Instantiating the gain decomposition.. Equation (2) can be evaluated exactly on the pooled eight-seed test records (20,880 utterance-seeds; 494 rescues, 384 harms, net +110 corrected labels, i.e. +0.53 accuracy points, matching the aggregate gain). The gate triggers on 26.5% of utterances. Within the triggered bucket the primary is wrong 56.5% of the time—a 1.8× enrichment over its unconditional error rate of 0.31, supplied purely by confidence ranking (the routing factor). On those triggered primary errors the reviewer disagrees 50.9% of the time (the independence factor; a homogeneous copy would score near zero here), and after the margin rule filters disagreements, 28.2% become flips, of which the reviewer is correct in 56.1% (the capability factor). The harm side shows why the margin is load-bearing: when the primary is in fact correct, the margin admits only 15.9% of disagreements as flips versus 28.2% on the error side, so potential harms are suppressed roughly twice as aggressively as potential rescues. Removing the margin ($m=0$) flips on every triggered disagreement and collapses the net gain, which is precisely the failure mode of the full-cost ensemble variants in Section 5.2.

A bias–variance interpretation.. The mechanism admits a natural reading through the lens of bias–variance decomposition. A fine-tuned LLM classifier has low variance on its own predictions (same input, same seed → same output) but high systematic bias on hard utterances (sarcasm, context-dependent polarity shifts). Same-model sampling (self-consistency, self-debate) averages over the model’s own output distribution, which reduces variance but cannot reduce bias—hence the uniformly negative results in Section 5.2. An independently trained reviewer,

even of the same capability tier, has a different bias profile because its pre-training corpus, tokenizer, and fine-tuning trajectory differ. When the gate routes only the low-confidence tail (where the primary’s bias is most likely to manifest), the reviewer’s independent bias has a chance to correct rather than reinforce the error. This explains why the gain is multiplicative in independence and capability: independence determines whether the reviewer’s bias is decorrelated; capability determines whether the decorrelated bias is more often correct than the primary’s on the routed subset.

Connection to ensemble theory.. Classical ensemble theory [13] identifies three conditions for ensemble improvement over a single model: accuracy, diversity, and a combination rule that exploits the diversity [49, 50]. Our setting differs from standard ensembles in two ways: (i) we combine at most two models, not a population; (ii) the combination rule is not uniform voting but a confidence-gated conditional substitution. The first difference means that the variance-reduction benefit of large ensembles is unavailable; the second means that we can exploit the primary model’s own uncertainty signal to decide when to consult the ensemble member, which large ensembles typically do not do (they vote on every input). The result is a mechanism that extracts most of the ensemble benefit at a fraction of the cost, at the price of a coverage-accuracy trade-off governed by τ .

Why confidence ranking, not calibration, suffices.. A natural objection is that the gate relies on confidence scores that are miscalibrated ($ECE \approx 0.1$; Section 5.1). Our resolution is that the gate requires only ranking quality—that low-confidence utterances are more likely to be wrong—not absolute calibration. This is a weaker requirement that is empirically well-satisfied (Spearman $\rho = 0.44$; AURC 0.156 vs. 0.307 for random). The isotonic recalibration experiment (Section 5.1) confirms that improving absolute calibration does not change the ranking, so the gate’s routing decisions are invariant to the miscalibration. This is fortunate because LLM confidence is notoriously difficult to calibrate absolutely, whereas ranking quality emerges more readily from the label-sequence posterior.

6.2. Why a coarse neutral specialist does not subsume the gate

Neutral is the plurality class on MELD (48% of test utterances), which suggests an obvious two-stage design: a first agent decides neutral versus emotional, and a second labels only the emotional cases. Our error audit explains both the appeal and the limit of this idea. Across seeds, 62.4% of the primary’s errors indeed straddle the neutral boundary: about 315 emotional utterances per test split are swallowed into neutral, 188 neutral utterances are false-alarmed as emotional, and 294 are confusions among emotional labels. Yet neutral is simultaneously the primary’s best-scored class (F1 81.1, versus 31.3 for fear), so

the residual difficulty is the boundary itself, not ignorance of the class.

Two offline tests, costing no additional training, bound what a coarse specialist could add. First, the independent reviewer—a different pre-trained family fine-tuned on the same task—already behaves like an independent neutral detector on these errors. It corrects only 20.0% of swallowed emotional utterances and assigns neutral to 67.8% of them, so roughly two thirds of the boundary blind spots are shared despite parameter-level independence; the false-alarm side is more recoverable (35.2% corrected), as are emotional-label confusions (18.0%). The swallowed cases concentrate in subdued emotions (83 joyful, 81 angry, and 73 sad utterances), i.e. subtle understatements rather than label confusion. Second, a deterministic repostulation of the same posterior cannot beat its own argmax: simulating the proposed cascade as a three-way router (confident neutral, confident emotion, otherwise escalate to the reviewer) and choosing both thresholds on the test set—an optimistic protocol—reaches at best 69.20 weighted-F1 at 1.24 $\times$ cost, below the frozen soft gate’s 69.37 at 1.27 $\times$. One-way vetoes (unidirectional neutral-to-emotion or emotion-to-neutral overrides) have optimistic ceilings of only +0.55 and +0.29 points. A genuine neutral-versus-emotion specialist can therefore earn its place only by introducing new information—a different training target, features, or context representation—rather than by restructuring the decision, and its headroom lies mainly on the false-alarm side. We treat such a class-conditional specialist, with thresholds calibrated on dev, as future work; under our frozen protocol it is not expected to replace soft gated arbitration.

6.3. Application value in deployment scenarios

As an illustrative projection—not measured on industrial data—consider a contact center processing 10^5 utterances per day. The eight-seed mean improvement of +0.49pp weighted-F1 (accuracy +0.53pp) corresponds to roughly 500 fewer mislabeled utterances per day under uniform class distribution; because ERC labels are highly imbalanced, the actual reduction depends on the class mix of the traffic. The price is a 1.27 $\times$ expected forward-pass budget (1.33 $\times$ in measured end-to-end latency): the two 7B models fit on a single 48 GB GPU (28.3 GB peak allocated, 34 GB reserved), batch-1 throughput remains 2.38 utterances/s under gating versus 3.16 for the primary alone, and 73% of traffic is answered by the primary model alone. In affective-monitoring settings where missed fear signals carry safety consequences, our per-class analysis advises the opposite of blind trust: flips on fear are unreliable (rescue/harm = 8/19, ratio 0.42), and the third branch of the framework—abstention to human review at very low confidence—is the appropriate handling, not forced arbitration (Section 5.5).

The cost-benefit profile is governed by two conditional probabilities that can be estimated from deployment logs. Let $P(\text{error} \mid \text{trigger})$ be the primary model’s error rate in

the low-confidence tail; on MELD this is $1 - 0.44 = 0.56$ (44% accuracy in the triggered bucket). Let $P(\text{rescue} | \text{error}, \text{trigger})$ be the probability that review corrects a triggered error; on MELD this is 0.48 (494 rescues out of 1024 triggered errors). The expected number of corrected errors per N utterances is then $N \times P(\text{trigger}) \times P(\text{error} | \text{trigger}) \times P(\text{rescue} | \text{error}, \text{trigger}) = N \times 0.267 \times 0.56 \times 0.48 \approx N \times 0.072$. For $N=10^5$, this is $\approx 7,200$ corrected errors per day at $1.27\times$ cost. The 500-utterance figure above is conservative because it does not condition on triggering; the full conditional estimate is higher but assumes the deployment distribution matches the test distribution, which is an extrapolation we mark explicitly.

6.4. Practical deployment guidelines

The control ladder and the decomposition of Section 3.3 translate into a short checklist for a practitioner considering gated review for a deployed ERC (or similar short-label) classifier.

Step 1: verify routing quality on the primary. Compute the Spearman correlation between the primary’s confidence and its correctness on a held-out split. If ranking quality is absent ($\rho \lesssim 0.2$), no gate can help, because the trigger set will not be enriched for errors; improve the primary or its confidence head first.

Step 2: verify reviewer independence. Measure the agreement rate between the candidate reviewer and the primary, restricted to the primary’s errors. Agreement above $\approx 90\%$ on errors (the homogeneous regime) means the reviewer shares the primary’s blind spots and review is wasted compute; this is the failure mode of self-debate and self-consistency. Independence is cheapest to obtain from a different pre-trained family; fine-tuning two checkpoints of the same base on the same data is not sufficient.

Step 3: verify capability parity on the routed subset, not globally. A reviewer that is weaker overall can still help if its accuracy on the low-confidence bucket exceeds the primary’s there; conversely, a reviewer that collapses on the routed distribution (the 1.5B case) produces net harm. The relevant comparison is conditional, and the routed subset is exactly where the primary is weakest, so parity on the full test set is neither necessary nor sufficient.

Step 4: calibrate the gate on dev, freeze, and report the frontier. The gate threshold is a one-dimensional parameter fitted on a development split; we recommend reporting the full τ sweep (the cost–accuracy frontier of Section 5.4) rather than a single operating point, since the deployment cost budget may differ from ours. All test numbers in this paper use thresholds frozen before any test evaluation.

Step 5: route rare or high-stakes classes to abstention, not to forced arbitration. Our per-class analysis (Section 5.7) shows the mechanism is net-negative on the two smallest classes; for classes where both models are unreliable, the third branch of the framework—abstention to human review—is the appropriate handling.

6.5. Why not a learned arbiter? Bounding the oracle gap

The oracle that picks the better of the two answers reaches 75.53, about six points above our rule gate (69.63). We probed this gap with a logistic arbiter trained on dev disagreements (223 examples; features: both confidences, their difference, low-confidence indicator, per-class priors) and evaluated once on test. The learned arbiter at its best threshold matches but does not exceed the rule gate (69.65 vs. 69.63), because the disagreement pool is intrinsically noisy: the reviewer is correct in only 33.6% of disagreements (179 potential rescues vs. 208 potential harms), and 27% of disagreements are both-wrong cases that no binary arbiter can exploit. The remaining oracle gap is therefore primarily a signal-to-noise constraint of the reviewer pair, not a deficiency of the arbitration rule; closing it requires either a third branch (abstention) or stronger base models rather than a cleverer gate.

7. Limitations

(i) Class-conditional failures. Fear-class flips are unreliable (2 rescues vs. 8 harms on 50 test instances), and surprise is also net-negative (32 rescues vs. 72 harms). High-stakes minority classes should be routed to the abstention branch rather than forced arbitration; the current implementation lacks a per-class gate, which would be a natural extension. (ii) Residual confounding in the family control. The Mistral reviewer matches the Qwen reviewer in standalone weighted-F1 (68.82 vs. 68.67) and routing rate (26.5% vs. 26.5%) but shares the LLaMA architectural lineage; lineage is bundled with tokeniser and optimisation similarities, so its null result does not isolate architecture alone. A factorial (capability $\times$ independence) sweep is future work that would tighten the attribution to either factor; the role-reversal symmetry itself rests on eight paired seeds in each direction (Section 5.3). (iii) Near-greedy self-consistency. Self-consistency was tested as a three-seed paired control at both $T=0.7$ and $T=0.3$ and is negative in 11 of 12 configurations; the remaining untested point is near-greedy temperature $T \rightarrow 0$, where sampling collapses toward greedy decoding and SC can add nothing by construction. (iv) Single-seed controls. Homogeneous self-debate and the weak-reviewer control remain single-seed (the τ -sweep Pareto uses three paired seeds, and the Mistral and role-reversal controls eight paired seeds each); multi-seed extension of the remaining single-run controls is future work. (v) Label-set asymmetry in cross-dataset transfer. The IEMOCAP transfer succeeds because its 4-class label set is a subset of MELD’s 7 classes (Section 5.3); a reverse transfer (from IEMOCAP-calibrated gate to MELD’s 7 classes) would be undefined, and any target label set with classes absent from the source set would require recalibration. (vi) Application-domain gap: both training corpora are scripted or performed English speech (TV sitcom and acted sessions), whereas the illustrative contact-center and affective-monitoring settings in-

volve ASR-transcribed spontaneous speech, different registers and noise, and different label taxonomies. A frozen zero-tuning transfer to spontaneous DailyDialog chat is negative (Section 5.3), so the cross-dataset evidence supports same-register transfer only; the deployment projections in Section 6.4 are extrapolations, and target-domain gate calibration plus reviewer verification are required before cross-register deployment claims.

8. Conclusion

We showed that a heterogeneous, comparably capable LLM acts as an effective selective verifier for conversational emotion recognition: invoked on the low-confidence quarter of traffic, it improves eight-seed weighted-F1 by +0.49 points ($p=0.019$) at $1.27\times$ expected cost, and the frozen gate transfers to IEMOCAP (+1.21, three seeds). A control ladder establishes that the gain requires independence, capability parity, and selective invocation jointly—same-model debate and self-consistency fail at two temperatures, while a weak reviewer fails despite independence. The arbitration rule nearly exhausts its available signal, and calibration analysis shows routing relies on confidence ranking rather than absolute values.

The contribution is a deployment-time mechanism orthogonal to training-time enhancements. Systems that reach 70+ weighted-F1 through retrieval, curriculum, or multimodal signals can still benefit from gated review, because the mechanism operates on any primary model with a usable confidence ranking, and the reviewer need only be comparably capable, not stronger. We verify this stackability directly by applying the gate to an independently trained InstructERC primary without touching its weights, obtaining a +1.33 gain after dev-level threshold recalibration; the practical checklist of Section 6.4 (routing quality, independence, capability parity on the routed subset, dev calibration, class-conditional abstention) gives a ready-to-use protocol for extending the mechanism to other short-label classification tasks.

Several directions remain. First, the gate threshold should be class-conditional, with separate τ per class to account for the systematic rescue-to-harm conversions on surprise and fear. Second, the oracle gap (75.53 vs. 69.63) is bounded by the reviewer’s intrinsic quality, not by the arbitration rule; closing it requires either a third model or a stronger base pair, not a cleverer gate. Third, cross-register transfer (DailyDialog) is negative, so target-domain calibration is required before deployment on spontaneous speech. Finally, an eight-seed role reversal—Qwen proposes, LLaMA2 reviews—independently replicates the gain (+0.71 $\pm$ 0.42, 8/8 seeds, $p=0.002$) with no significant difference between pairing directions, and the InstructERC proponent-stack experiment shows the mechanism is also portable across primary training pipelines, but only after dev-level recalibration of the gate: the frozen thresholds encode the primary’s confidence semantics and do not transfer verbatim (Section 5.3). A systematic multi-seed

verification across further external pipelines remains future work.

Appendix A. Prompt and truncation protocol

Classification prompt.. The prompt lists the admissible label words, the speaker-tagged dialogue history (one turn per line), and the target utterance, and instructs the model to answer with exactly one label word from the list and nothing else. The reviewer receives the identical contract; its optional hint appends a three-item self-check (literal wording versus contextual affect; speaker-state continuation versus reaction; degeneration toward majority classes on rare labels). Both prompt variants are released verbatim in the code archive.

Left truncation.. All prompts are tokenized with the tokenizer’s automatic truncation disabled. When the sequence exceeds the model context, history is truncated from the left while the BOS token, the target utterance, and the most recent turns are retained, so the tail semantics required for label generation are never cut.

Generation and scoring budget.. Labels are generated greedily with `max_new_tokens=8` (sufficient for one label word plus a newline). Confidence scoring adds exactly one teacher-forced forward pass per utterance over the candidate label sequences; no long-form rationale is generated at deployment.

Appendix B. DailyDialog label mapping and decoding protocols

DailyDialog’s seven emotion labels are mapped onto the MELD label set by renaming only two classes and keeping the other five identical (Table B.7); after the mapping the label schema is exactly the MELD schema, so the MELD-trained models are applied without any architectural or prompt change. Because DailyDialog provides no speaker annotations, speakers are assigned by alternating turns (A/B), matching the history format used for MELD.

Table B.7: DailyDialog $\rightarrow$ MELD label mapping (7 $\rightarrow$ 7).

DailyDialog label	MELD label
no emotion	neutral
happiness	joyful
anger	anger
disgust	disgust
fear	fear
sadness	sadness
surprise	surprise

Two decoding protocols are reported in Section 5.3. Greedy: the label is the first generated word, the protocol used throughout this paper. Logprob-argmax: each candidate label sequence is teacher-forced and scored by

its mean token log-probability, and the highest-scoring label is returned—the same quantity that defines the gating confidence p_a (Section 3.2). The gated DailyDialog result uses logprob-argmax so that the baseline prediction and the gate confidence come from one decoding family; the greedy number is given as a reference for comparability with the rest of the paper. The two protocols can differ substantially under the extreme class skew of DailyDialog (81.7% neutral utterances in the test split), which is one reason we treat this cross-register evaluation as preliminary.

CRedit authorship contribution statement

First Author: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing—original draft, Visualization. Second Author: Supervision, Writing—review & editing, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

MELD and IEMOCAP are publicly available benchmark datasets (MELD: <https://github.com/declare-lab/MELD>; IEMOCAP: available upon request from USC SAIL). All prompts, evaluation scripts, per-seed prediction records, prompts, evaluation and aggregation scripts, per-utterance prediction records for all seeds, and trained LoRA adapters are archived in a permanent public repository with a Zenodo DOI (DOI inserted at submission). The LLaMA2-7B base weights remain subject to Meta’s license and must be obtained from Meta; all training scripts needed to reproduce the adapters are included.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large-language-model coding assistant for drafting LaTeX boilerplate and figure scripts. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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